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A Preliminary Tomography Inversion Study on the Palu Koro Fault, Central Sulawesi Using BMKG Seismic Network

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Abstract. The Palu Koro Fault, which frequently causes solid and destructive earthquakes, is the most exciting aspect of Central Sulawesi's tectonics. We used local earthquake data from the area in this preliminary study. Our preliminary investigation entails analyzing local seismic data and estimating the velocity structure and hypocenter location using a tomographic approach that employs P-wave arrival times from the Agency for Meteorology, Climatology, and Geophysics earthquake observation network from 2010 to 2019. We use a 1-D initial velocity structural model in this tomographic inversion method, and the results are evaluated using the checkerboard test, derivative weight sum, and ray hit count. To the west and east of the Palu Koro Fault area, preliminary tomographic inversion results show a relatively high seismic velocity. These areas are interpreted as West Sulawesi Mandala and East Sulawesi Mandala Poso Fault and Wekuli Fault. The low-velocity anomaly is found near the Palu Koro Fault. Because of the high-speed thrusting, the relocated hypocenters also clustered around the Palu Koro Fault area. In the following activity, we will determine the speed structure of the S wave, which is expected to provide better geological information in the interpretation.

Keywords: Palu Koro Fault, Local earthquakes, Tomography inversion, BMKG

1. Introduction

Seismic patterns on the island of Sulawesi are always influenced by tectonic activities below the earth's surface, such as faults, troughs, trusts, and spreading centers in the Makassar Strait. Of the several activities on the island of Sulawesi, the Palu Koro Fault is the dominant one as a tectonic source of the emergence of natural disasters in Central Sulawesi [1–4]. This fault is located between several large plates, so it has the potential to cause very high earthquakes and tsunamis [5–7].

The seismicity distribution on the Sulawesi island shows that the Sulawesi region has many earthquakes, spread over all areas with a magnitude of 5 Mw with shallow to deep depths (Figure 1).

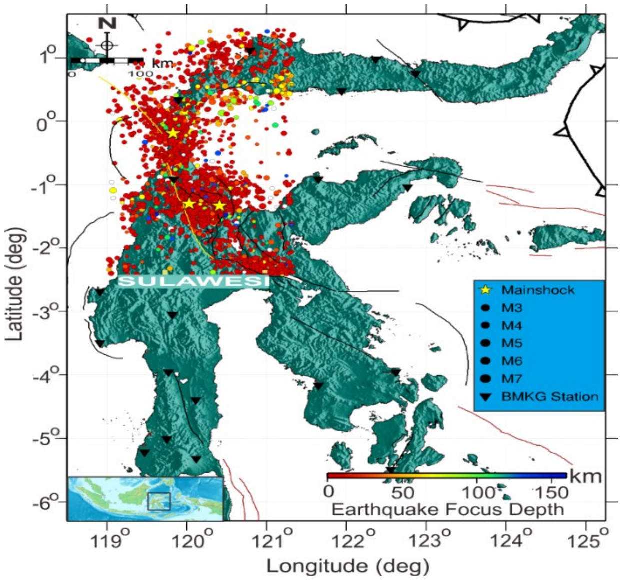


Figure 1. Overview of Sulawesi's seismicity (Indonesia) earthquake magnitude 3 Mw. It can be seen that there were 7.5 Mw earthquakes in 2018 in Palu (Central Sulawesi), 6.6 Mw in 2017 in Poso (Central Sulawesi), and 6.2 Mw in 2012 in Kulawi (Central Sulawesi) which are marked by a yellow star. The black inverted triangle depicts earthquake recording stations in all Sulawesi areas. This data was obtained from the BMKG earthquake catalog data from 2010-2019.

Therefore, by observing the level of seismic activity in Sulawesi, especially in the Palu Koro Fault zone, which once caused the natural tsunami disaster, it is necessary to conduct a study that can identify subsurface structure models that will later be used in the preparation of disaster mitigation strategies and can study the existence the Palu Koro fault and other faults in the Central Sulawesi region. Studies on seismicity in the Palu Koro Fault zone have previously been carried out by [8–11] regarding the relocation of the hypocenter of September 28, 2018, 7.5 Mw earthquake, mapping of fault lines on the coast, seismic activity before the Palu earthquake and the level of coulomb stress along the Palu Koro Fault.

A seismic tomographic study of travel time with local data around the Palu Koro fault zone was carried out to image the subsurface structure model. This study uses travel time seismic tomography, which will analyze the 3-D seismic velocity structure of the P wave in the resulting tomogram to describe the seismicity and tectonic patterns around the Palu Koro Fault zone for use in disaster mitigation planning.

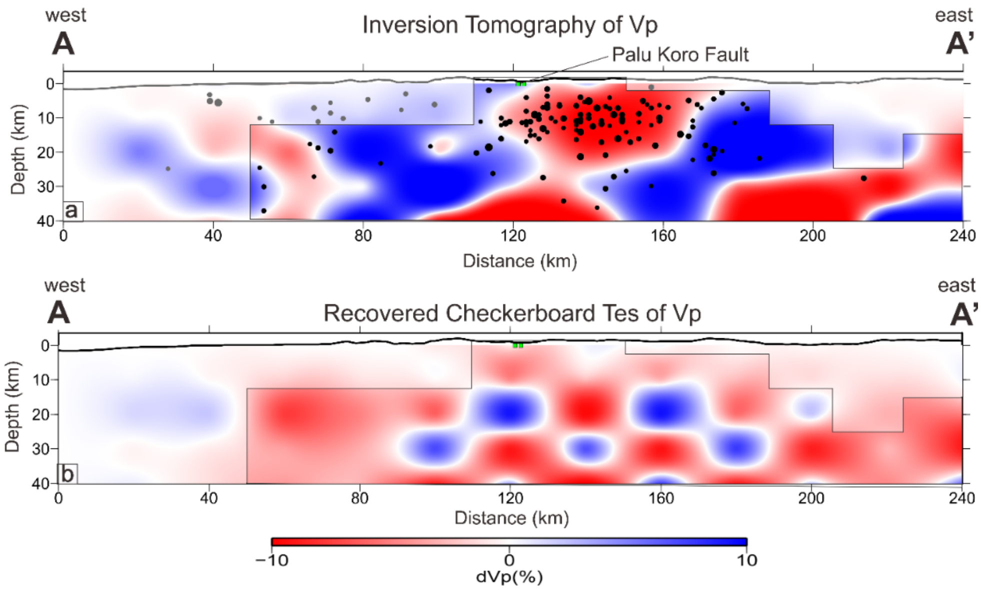
2. Data and Method

In this study, we use data from the Agency for Meteorology, Climatology, and Geophysics (BMKG) station network in the form of local earthquake arrival time data to determine the Vp velocity structure and the same time to relocate the hypocenter by applying seismic tomography inversion. The data recorded by 20 BMKG seismic station in the Sulawesi island area, with a recording period from January 1, 2010, to December 31, 2019. The total number of earthquakes that occurred was 2.850, with a P-wave phase of 22.319 with depths of 1 to 160 km and magnitudes of 3 to 7.4 Mw. In the analysis, we ensure that the station records 6 phases for use in the seismic tomography process. We applied SIMULPS12 [12] to calculate the velocity structure values of Vp and Vp/Vs, along with the determination of hypocenter relocation. The travel time from the earthquake source to the receiver for 3D velocity through the beam path is calculated using the pseudo bending technique [13] . In this study, we used the velocity of the incident P wave to determine its velocity structure. The initial velocity model used in this seismic tomography is the 1-D AK135 initial velocity structure [14]. To obtain an excellent seismic tomography resolution, we analyzed using a checkerboard resolution test, diagonal resolution element, derivative weight sum, and ray hit count [15]. We used a dense horizontal grid node of 20 x 20 km^2 and an average for a vertical grid node of 10 km based on data on earthquake distribution, station distribution, and average ray traces [16–18].

3. Result and Discussion

We established the P-wave velocity structure in the region around the Palu Koro Fault using local network data from BMKG stations. The initial results are a low anomaly (red color) around the Palu Koro Fault area (Figure 2a), located at a depth of 5-25 km. This is by the research results for the relocation of earthquake hypocenters from 2009 to 2018, showing that earthquakes around the Palu Koro and Matano fault areas had earthquake characteristics with a depth of 20 km [8]. Another feature that appears in the presence of a high anomaly (blue color) coincides with a low anomaly. This is indicated as a geological structure formed by different generations in the form of the West Sulawesi Mandala (Mamuju fault) and the East Sulawesi Mandala, which consists of the Poso Fault and the Wekuli Fault [19]. In the profile of the miners from the vertical seismic tomography, it can be seen that the earthquake relocation results were in the high anomaly zone. Some were gathered in the weak zone anomaly, interpreted as the Palu Koro fault area. The results of this seismic tomography inversion are supported by good ray path reversal in the test checkerboard profile (Figure 2b), diagonal resolution element values (Figure 2c), and derivative weight sum (Figure 2d), and ray hit count (Figure 2e).

The initial results of this seismic tomographic inversion are expected to provide new insights into geology and geodynamic interpretation, especially for active fault zones related to seismicity in the study area.



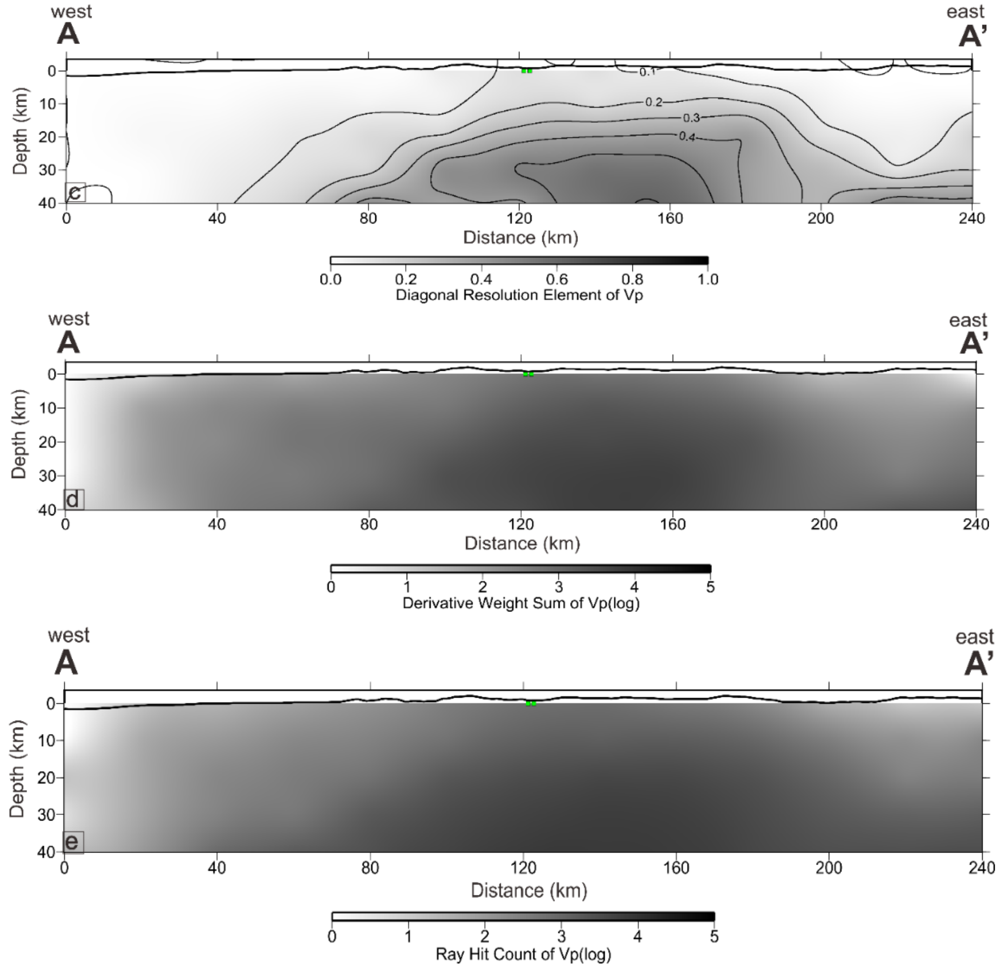


Figure 2. Cross section on one of the tracks that pass through the Palu Koro Fault zone. (a) The results of the vertical cross-section of the seismic tomography inversion for the Vp structure with a west-east direction (C-C'); (b) the results of the vertical cross-section for the checkerboard test; (c) the result of the vertical cross-section for diagonal resolution element; (d) the results of the vertical cross-section for the derivative weight sum; (e) result of the vertical cross-section for ray hit count. The cross-section's red and blue colors indicate low and high anomalies. The derivative weight sum and ray hit count indicate their respective values for the gray color in the diagonal resolution element section. The white circle in Figure 2a is the result of the hypocenter's relocation, and the red squares in each section are projections of the position of the Palu Koro Fault.

4. Conclusions

We have carried out the results of an interim study to image subsurface structures with Vp seismic tomography around the active Palu Koro Fault zone. This study found that the resolution test was quite good around the Palu Koro Fault zone with a ray path reversal at a depth of 10 – 30 km and several other resolution tests. The tomogram from the vertical tomography inversion shows a negative (low) anomaly around the Palu Koro Fault zone at a depth of 5-25 km. This can be interpreted that the geological structure in the study area being a sedimentary layer that has undergone an overhaul and is found at shallow depths with high rock porosity. There is also a positive (high) anomaly between the weak zones, which is indicated as an ultra-alkaline zone where the area is a different geological structure in the West Sulawesi Mandala in the Mamuju area and the East Sulawesi Mandala located in the Poso area.

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